We will begin at 9:05 a.m.

Lecture 23: Vectors

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2024/2025

Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Define a vector in 2D and 3D, including its magnitude, direction, and standard position.
- ☐ Determine the component form of a vector given initial and terminal points.
- ☐ Calculate the magnitude of a vector
- ☐ Perform vector operations and apply vector operations properties.
- ☐ Express a vector as a linear combination of the standard unit vectors.
- ☐ Find a unit vector in the direction of a given vector.
- ☐ Compute the midpoint of a line segment between two points in space.

Introduction

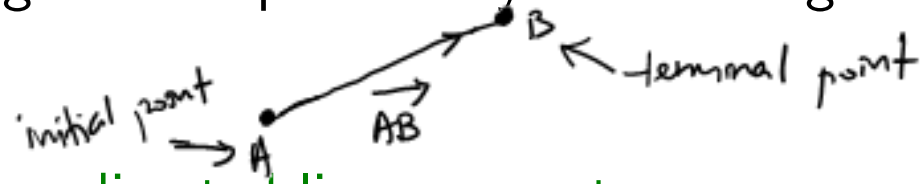
There are two types of quantities we measure

- scalar – measured only by the magnitude.
  e.g. mass, length, time, temperature
- vector – measured by both magnitude and direction
  e.g. displacement, velocity, accerelation, force.



Vectors

A vector quantity is anything that is specified by both a magnitude and a direction.



The vector is represented by a directed line segment.

The directed line segment $\overrightarrow{AB}$ has initial point A and terminal point B and its length is denoted by $\left|\overrightarrow{AB}\right|$.



$\overrightarrow{AB} \parallel \overrightarrow{CD}$, $|\overrightarrow{AB}| = |\overrightarrow{CD}|$ $\Rightarrow \overrightarrow{AB} = \overrightarrow{CD}$

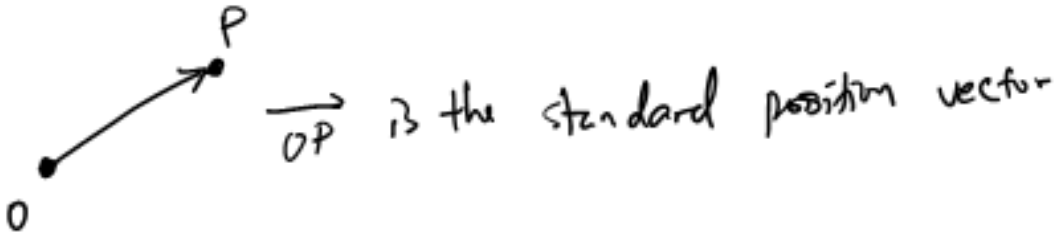
Two vectors are equal if they have the same length and direction.

Notation: $\boldsymbol{v}$, $\vec{v}$, $\overrightarrow{PQ}$, $\underset{\sim}{v}$

$\overrightarrow{PQ} \neq \overrightarrow{QP}$ instead $\overrightarrow{PQ} = -\overrightarrow{QP}$

bolded

Standard position vector is the line segment with the initial point at the origin.



Vectors

If $\boldsymbol{v}$ is a two-dimensional vector in the plane equal to the vector with initial point at the origin $(0,0)$ and terminal point (v_1, v_2), then the component form of $\boldsymbol{v}$ is

$$\boldsymbol{v} = \langle v_1, v_2 \rangle.$$

If $\boldsymbol{v}$ is a three-dimensional vector equal to the vector with initial point at the origin and terminal point (v_1, v_2, v_3), then the component form of $\boldsymbol{v}$ is

$$\boldsymbol{v} = \langle v_1, v_2, v_3 \rangle.$$

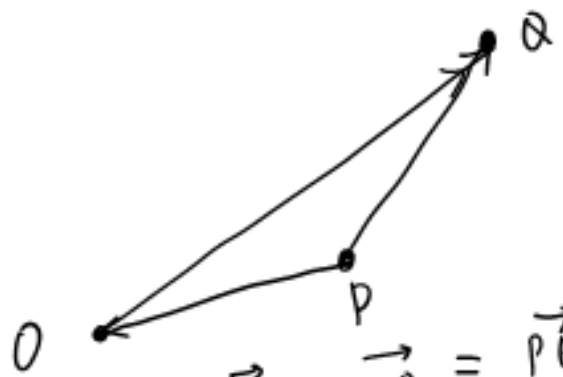
If the vector $\boldsymbol{v} = \langle v_1, v_2, v_3 \rangle$ is represented by the directed line segment $\overrightarrow{PQ}$, where $P(x_1, y_1, z_1)$ is the initial point and $Q(x_2, y_2, z_2)$ is the terminal point, then

$$\langle v_1, v_2, v_3 \rangle = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle.$$

Eg: $P(1,2,3)$, $Q(4,5,6)$

$\overrightarrow{PQ} = \langle 4-1,\ 5-2,\ 6-3 \rangle = \langle 3, 3, 3 \rangle$

$\overrightarrow{QP} = \langle 1-4,\ 2-5,\ 3-6 \rangle = \langle -3, -3, -3 \rangle = -\overrightarrow{PQ}$

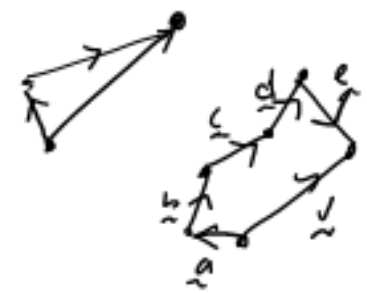


$\overrightarrow{PO} + \overrightarrow{OQ} = \overrightarrow{PQ}$

$-\overrightarrow{OP} + \overrightarrow{OQ} = \overrightarrow{PQ}$

$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$ ← standard position vector of P

↑ standard position vector of Q



$\underset{\sim}{v} = \underset{\sim}{a} + \underset{\sim}{b} + \underset{\sim}{c} + \underset{\sim}{d} + \underset{\sim}{e}$

Length

The magnitude (or length) of the vector
$\boldsymbol{v} = \langle v_1, v_2, v_3 \rangle = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$ is the nonnegative number

$$|\boldsymbol{v}| = \sqrt{{v_1}^2 + {v_2}^2 + {v_3}^2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

Example: Find the component form and length of the vector with initial point $P(-3, 4, 1)$ and terminal point $Q(-5, 2, 2)$.

$\overrightarrow{PQ} = \langle -5-(-3), 2-4, 2-1 \rangle$

$= \langle -2, -2, 1 \rangle$ ← component form of the vector

$|\overrightarrow{PQ}| = \sqrt{(-2)^2 + (-2)^2 + (1)^2}$

$= \sqrt{4+4+1} = 3$ ← length of vector $\overrightarrow{PQ}$

* zero vector $\vec{0}$ is the only vector with length 0 and have no specific direction.. (we stay at the point)

Vector Algebra Operations

Let $\boldsymbol{u} = \langle u_1, u_2, u_3 \rangle$ and $\boldsymbol{v} = \langle v_1, v_2, v_3 \rangle$ be vectors with k a scalar.

add each of the respective component

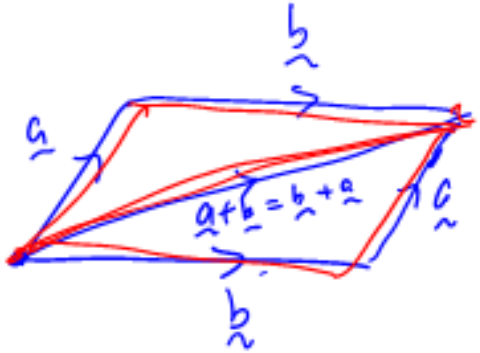
Addition: $\boldsymbol{u} + \boldsymbol{v} = \langle u_1 + v_1, u_2 + v_2, u_3 + v_3 \rangle$

Scalar multiplication: $k\boldsymbol{u} = \langle ku_1, ku_2, ku_3 \rangle$ multiply the scale. to each of the component.

Difference: $\boldsymbol{u} - \boldsymbol{v} = \boldsymbol{u} + (-\boldsymbol{v}) = \langle u_1 - v_1, u_2 - v_2, u_3 - v_3 \rangle$

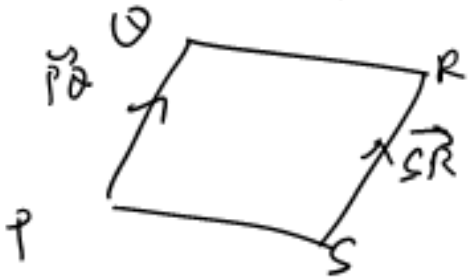
Vector addition

① Parallelogram Law

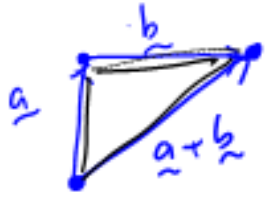


A quadrilateral PQRS is a parallelogram iff $\overrightarrow{PQ} = \overrightarrow{SR}$

PQ and SR have the same length and are parallel.



Triangle Law



$|\underset{\sim}{a}+\underset{\sim}{b}| \leq |\underset{\sim}{a}| + |\underset{\sim}{b}|$, the equality holds when $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are colinear (on a same line)

Scalar multiplication

Two vectors are parallel iff one is the scalar multiple of the others. (i.e. $\underset{\sim}{u}$ is parallel to $\underset{\sim}{v}$ iff $\underset{\sim}{v} = c\underset{\sim}{u}$ for some scalar c.)

For example: Determine whether $\underset{\sim}{u} = \langle 1, 2, 3 \rangle$, $\underset{\sim}{v} = \langle -2, -4, -6 \rangle$

$\underset{\sim}{v} = \underset{\text{-ve}}{-2}\,\underset{\sim}{u}$, so $\underset{\sim}{u}$ and $\underset{\sim}{v}$ are parallel and they are in the opposite direction.

Vector Algebra Operations

Example: Let $\boldsymbol{u} = \langle -1, 3, 1 \rangle$ and $\boldsymbol{v} = \langle 4, 7, 0 \rangle$. Find the components of

(a) $2\boldsymbol{u} + 3\boldsymbol{v}$

(b) $\boldsymbol{u} - \boldsymbol{v}$

(c) $\left|\frac{1}{2}\boldsymbol{u}\right|$

(a) $2\underset{\sim}{u} + 3\underset{\sim}{v} = 2\langle -1, 3, 1 \rangle + 3\langle 4, 7, 0 \rangle$

$= \langle -2, 6, 2 \rangle + \langle 12, 21, 0 \rangle$

$= \langle 10, 27, 2 \rangle$

(b) $\underset{\sim}{u} - \underset{\sim}{v} = \langle -1, 3, 1 \rangle - \langle 4, 7, 0 \rangle$

$= \langle -5, -4, 1 \rangle$

(c) $\left|\frac{1}{2}\underset{\sim}{u}\right| = \left|\frac{1}{2}\langle -1, 3, 1 \rangle\right|$

$= \left|\left\langle -\frac{1}{2}, \frac{3}{2}, \frac{1}{2} \right\rangle\right|$

$= \sqrt{\left(-\frac{1}{2}\right)^2 + \left(\frac{3}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \frac{\sqrt{11}}{2}$.

Properties of Vector Operations

Let $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ be vectors and a, b be scalars.

1. $\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$ commutative in addition.
2. $(\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w} = \boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w})$ associative in addition.
3. $\boldsymbol{u} + \boldsymbol{0} = \boldsymbol{u}$ Existence of addition identity. 0 (zero vector)
4. $\boldsymbol{u} + (-\boldsymbol{u}) = \boldsymbol{0}$ Existence of addition inverse. (negative)
5. $0\boldsymbol{u} = \boldsymbol{0}$
6. $1\boldsymbol{u} = \boldsymbol{u}$
7. $a(b\boldsymbol{u}) = (ab)\boldsymbol{u}$ Associative in scalar multiplication.
8. $a(\boldsymbol{u} + \boldsymbol{v}) = a\boldsymbol{u} + a\boldsymbol{v}$ } Distributive Law.
9. $(a + b)\boldsymbol{u} = a\boldsymbol{u} + b\boldsymbol{u}$

on the same line – colinear

If three or more space vectors lie in the same plane, we say they are coplanar vectors

Unit Vector

A vector $\boldsymbol{v}$ of length 1 is called a unit vector.

The standard unit vectors are

$$\boldsymbol{i} = \langle 1, 0, 0 \rangle, \quad \boldsymbol{j} = \langle 0, 1, 0 \rangle, \quad \boldsymbol{k} = \langle 0, 0, 1 \rangle.$$

$|\underset{\sim}{i}| = \sqrt{1^2 + 0^2 + 0^2} = 1$ $\quad |\underset{\sim}{j}| = 1$ $\quad |\underset{\sim}{k}| = 1$

Any vector $\boldsymbol{v} = \langle v_1, v_2, v_3 \rangle$ can be written as a linear combination of the standard unit vectors.

$$\underset{\sim}{v} = \langle v_1, v_2, v_3 \rangle = \langle v_1, 0, 0 \rangle + \langle 0, v_2, 0 \rangle + \langle 0, 0, v_3 \rangle$$

$$= v_1 \langle 1, 0, 0 \rangle + v_2 \langle 0, 1, 0 \rangle + v_3 \langle 0, 0, 1 \rangle$$

$$= v_1 \underset{\sim}{i} + v_2 \underset{\sim}{j} + v_3 \underset{\sim}{k}$$

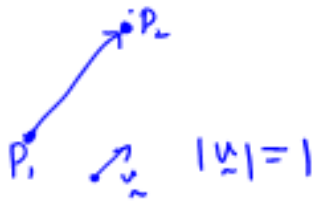
standard unit vectors

Unit Vector

If $\boldsymbol{v} \neq \mathbf{0}$, then

1. $\frac{\boldsymbol{v}}{|\boldsymbol{v}|}$ is a unit vector called the direction of $\boldsymbol{v}$;
2. the equation $\boldsymbol{v} = |\boldsymbol{v}|\frac{\boldsymbol{v}}{|\boldsymbol{v}|}$ expresses $\boldsymbol{v}$ as its length times its direction.

Example: Find a unit vector $\boldsymbol{u}$ in the direction of the vector from $P_1(1, 0, 1)$ to $P_2(3, 2, 0)$.



Let $\underset{\sim}{v} = \overrightarrow{P_1P_2} = \langle 3-1, 2-0, 0-1 \rangle = \langle 2, 2, -1 \rangle$

$|\underset{\sim}{v}| = \sqrt{2^2 + 2^2 + (-1)^2} = 3$

$|\underset{\sim}{u}| = \frac{\underset{\sim}{v}}{|\underset{\sim}{v}|} = \frac{\langle 2, 2, -1 \rangle}{3} = \langle \frac{2}{3}, \frac{2}{3}, \frac{-1}{3} \rangle$

Check: $|\underset{\sim}{u}| = \sqrt{(\frac{2}{3})^2 + (\frac{2}{3})^2 + (\frac{-1}{3})^2} = 1$

Unit Vector

Example: Express $\boldsymbol{v} = 3\boldsymbol{i} - 4\boldsymbol{j}$ as a product of its magnitude times its direction.

$$\underset{\sim}{v} = |\underset{\sim}{v}| \frac{\underset{\sim}{v}}{|\underset{\sim}{v}|}$$

magnitude · direction.

$$|\underset{\sim}{v}| = \sqrt{3^2 + (-4)^2} = 5$$

$$\underset{\sim}{v} = |\underset{\sim}{v}| \cdot \frac{\underset{\sim}{v}}{|\underset{\sim}{v}|} = 5 \left(\frac{3\underset{\sim}{i} - 4\underset{\sim}{j}}{5} \right)$$

$$= 5 \left(\frac{3}{5}\underset{\sim}{i} - \frac{4}{5}\underset{\sim}{j} \right)$$

magnitude · direction.

Unit Vector

Example: Find a vector of magnitude 6, in the direction of the vector $\boldsymbol{v} = 2\boldsymbol{i} + 2\boldsymbol{j} - \boldsymbol{k}$.

Let $\underset{\sim}{u}$ be the vector we are calculating

magnitude : $|\underset{\sim}{u}| = 6$

direction : $\dfrac{\underset{\sim}{v}}{|\underset{\sim}{v}|} = \dfrac{2\underset{\sim}{i} + 2\underset{\sim}{j} - \underset{\sim}{k}}{\sqrt{(2)^2 + (2)^2 + (-1)^2}} = \dfrac{2\underset{\sim}{i} + 2\underset{\sim}{j} - \underset{\sim}{k}}{3}$

$= \frac{2}{3}\underset{\sim}{i} + \frac{2}{3}\underset{\sim}{j} - \frac{1}{3}\underset{\sim}{k}$

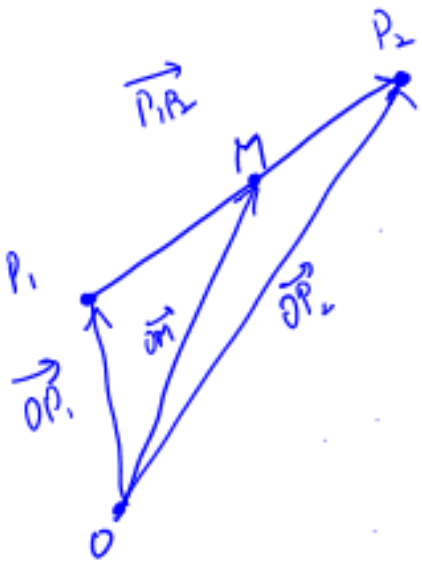
$\underset{\sim}{u} = |\underset{\sim}{u}| \cdot \dfrac{\underset{\sim}{v}}{|\underset{\sim}{v}|} = 6\left(\frac{2}{3}\underset{\sim}{i} + \frac{2}{3}\underset{\sim}{j} - \frac{1}{3}\underset{\sim}{k}\right)$

$= 4\underset{\sim}{i} + 4\underset{\sim}{j} - 2\underset{\sim}{k}$

Midpoint of a Line Segment

The midpoint M of the line segment joining points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is the point

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right).$$



$\overrightarrow{OP_1} = \langle x_1, y_1, z_1 \rangle$ $\overrightarrow{OP_2} = \langle x_2, y_2, z_2 \rangle$

To find the coordinate of M, we first need to know the standard position vector $\overrightarrow{OM}$

$\overrightarrow{OM} = \overrightarrow{OP_1} + \overrightarrow{P_1M}$

M is the midpoint of P_1P_2

$\overrightarrow{P_1M} = \frac{1}{2}\overrightarrow{P_1P_2}$

$= \overrightarrow{OP_1} + \frac{1}{2}\overrightarrow{P_1P_2}$

$= \langle x_1, y_1, z_1 \rangle + \frac{1}{2}\langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$

$$= \left\langle \frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2} \right\rangle$$

So the coordinate of M is $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2} \right)$

$\langle 1, 1, 1 \rangle$ is a vector

$(1, 1, 1)$ is a coordinate

Midpoint of a Line Segment

Example: Find the midpoint of the segment joining $P_1(3, -2, 0)$ and $P_2(7, 4, 4)$.

$$\left(\frac{3+7}{2}, \frac{-2+4}{2}, \frac{0+4}{2}\right) = (5, 1, 2)$$